

CUSTOMER SEGMENTATION AND CHURN PATTERN ANALYTICS IN EUROPEAN BANKING

Data Science Internship Research Paper

Prepared By : Baikan Roshini

Internship : Data Science

Technologies : Python, Pandas, NumPy, Plotly and Streamlit

Abstract

This study investigates customer churn and customer segmentation patterns in a European banking dataset containing 10,000 customer records and 14 variables covering customer demographics, credit profile, tenure, account balance, product usage, credit-card ownership, activity, estimated salary and churn status. The analysis focuses on demographic, geographic, financial, product-usage and engagement characteristics associated with customer attrition. A structured data-preparation and exploratory-analytics workflow was implemented using Python, Pandas and NumPy, followed by feature engineering to create age groups, credit-score bands, tenure groups, balance segments, customer-value categories and engagement labels. The observed overall churn rate was 20.37%, with 2,037 customers classified as churned. Germany showed the highest geographic churn rate at 32.44%, while the 46–60 age group recorded the highest age-segment churn rate at 51.12%. Inactive customers also showed a higher churn rate (26.85%) than active customers (14.27%). High-value customers represented 2,516 records and had a churn rate of 25.08%. The findings are presented through an interactive Streamlit dashboard with dynamic KPIs, filters and visualizations. The study concludes that churn is not evenly distributed across the customer base and that segment-aware retention strategies can support more focused customer-management decisions.

Keywords: Customer Churn, Customer Segmentation, Banking Analytics, Exploratory Data Analysis, Feature Engineering, Streamlit, Customer Retention

1. Introduction

Customer retention is a major analytical concern for retail banking because existing customers represent established relationships, transaction histories and opportunities for continued product engagement. Customer churn, commonly defined as a customer leaving a service relationship, can therefore create a need for costly replacement acquisition and can weaken long-term customer value. Data-driven churn analysis provides a systematic way to identify differences between retained and churned customers and to prioritize segments for further investigation.

Recent banking research has combined exploratory analysis, machine learning and visualization to study customer churn. For example, recent work has demonstrated the use of machine-learning models and visualization applications for banking churn analysis, while other studies have emphasized the value of demographic, financial and behavioral variables in churn modeling. These studies support the use of structured customer-level data as a basis for retention analysis.

This project takes a deliberately analytics-focused approach. Rather than claiming a predictive production model, it examines observed churn patterns and constructs interpretable customer segments that can be explored interactively. The output is a Streamlit application intended to support exploratory business decision-making.

2. Problem Definition and Research Objectives

The central research problem is to determine how customer churn varies across identifiable demographic, geographic, financial, product and engagement segments and how these patterns can be presented in an actionable interactive dashboard.

Research question:

How can customer-level banking data be transformed into interpretable segments that reveal differences in churn behavior and support targeted retention analysis?

Objectives:

- Validate and preprocess the European banking customer dataset.
- Explore distributions and relationships among customer characteristics and churn.
- Engineer interpretable demographic, financial, tenure and engagement segments.
- Compare churn rates across geographic, demographic and behavioral groups.
- Identify high-value customer segments with elevated churn exposure.
- Develop an interactive Streamlit dashboard for segment-level exploration.
- Translate observed patterns into practical retention recommendations while distinguishing association from causation.

3. Literature Review

Banking churn research commonly treats customer attrition as a classification or customer-management problem. Studies have applied logistic regression, support vector machines, decision trees, random forests, gradient boosting and ensemble methods to identify customers at risk of leaving. More recent work has also integrated explainability, class-imbalance techniques and interactive visualization.

The literature also supports customer segmentation as a complementary analytical perspective. Segmentation can organize heterogeneous customers into interpretable groups, allowing banks to distinguish between customer profiles rather than applying a single retention strategy to the entire population. Recent banking segmentation research has used clustering methods such as K-means and DBSCAN to create actionable customer groups.

The present project differs from predictive churn studies because its primary contribution is descriptive and diagnostic. The purpose is to expose segment-level churn patterns through transparent engineered categories and an interactive dashboard. This approach is appropriate for an internship-level business analytics application where interpretability and usability are important alongside technical implementation.

4. Methodology

4.1 Data Source and Structure

The study uses the uploaded `European_Bank.csv` dataset. The dataset contains 10,000 customer observations and 14 original columns, including a year field, customer identifier, surname, credit score, geography, gender, age, tenure, balance, number of products, credit-card ownership, active-member status, estimated salary and churn indicator. The target variable is `Exited`, where 1 represents a churned customer and 0 represents a retained customer.

For the dashboard, the surname field is not used for analysis because it does not provide a meaningful behavioral or financial segmentation variable. Customer identifiers are also treated as identifiers rather than analytical predictors.

4.2 Data Cleaning and Pre-processing

- Duplicate customer records were removed before analysis.
- Required numerical variables were treated as numeric fields.
- Categorical variables such as Geography and Gender were retained for grouped analysis.
- The target variable `Exited` was used to distinguish churned and retained customers.

- The preprocessing pipeline was designed to preserve the original customer-level observations while adding analytical segment labels.

4.3 Exploratory Data Analysis

Exploratory analysis was performed using univariate and bivariate summaries. Churn rates were calculated as the proportion of customers with `Exited = 1` within each segment. This rate-based approach is preferable to comparing raw churn counts alone because segment sizes differ.

4.4 Feature Engineering and Segmentation

Engineered Feature	Categories	Analytical Purpose
Age Group	<30; 30–45; 46–60; 60+	Compare churn across age bands
Credit Score Band	Poor; Fair; Good; Very Good; Excellent	Compare financial-risk bands
Tenure Group	New; Mid-term; Established; Long-term	Compare relationship duration
Balance Segment	Zero; Low; Medium; High; Very High	Compare financial balance profiles
Customer Value	High Value; Standard Value	Identify financially important segments
Engagement	Active; Inactive	Compare customer activity
Churn Status	Churned; Retained	Create interpretable target labels

The High Value label is an analytical portfolio segmentation rather than a financial valuation model. In this project, a customer is classified as High Value when both Balance and EstimatedSalary are at or above their respective dataset medians.

5. Results and Findings

5.1 Overall Churn Profile

The dataset contains 10,000 customers, of whom 2,037 are classified as churned and 7,963 as retained. The overall churn rate is 20.37%. The average account balance is 76,485.89, and approximately 51.51% of customers are classified as active members.

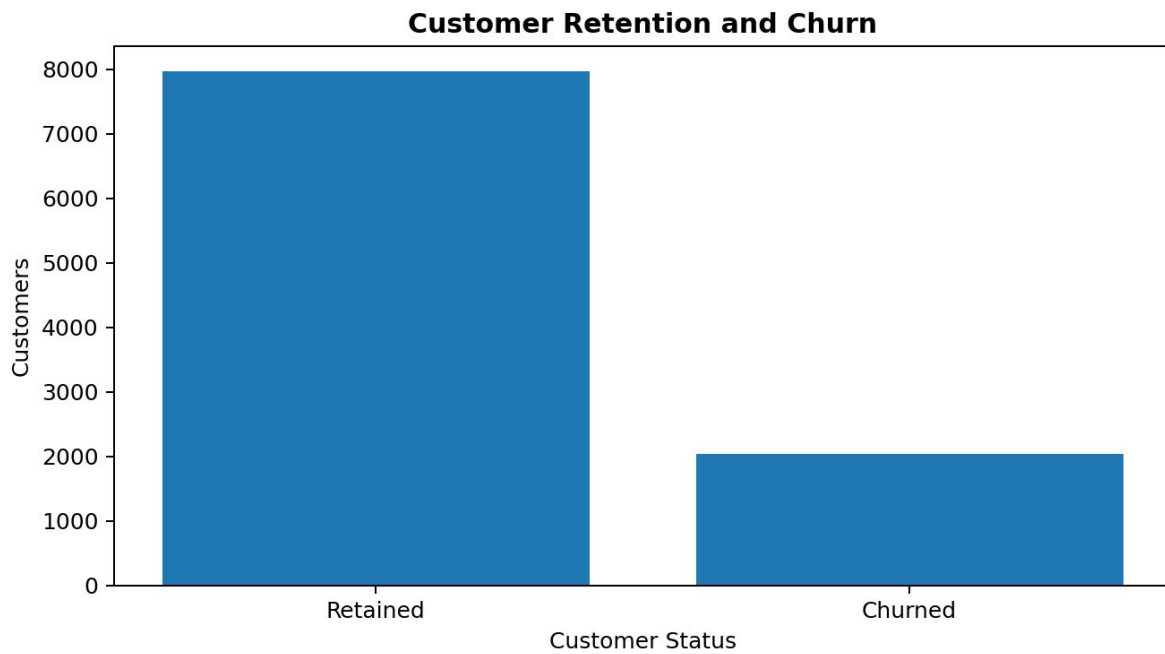


Figure 1. Overall retained versus churned customer counts.

5.2 Geographic Churn Patterns

Germany contains 2,509 customers and has an observed churn rate of 32.44%.

Spain contains 2,477 customers and has an observed churn rate of 16.67%.

France contains 5,014 customers and has an observed churn rate of 16.15%.

Germany records the highest churn rate among the three geographic groups at 32.44%, substantially above France and Spain. This difference identifies geography as a useful segmentation dimension for further retention investigation, although the analysis does not establish that geography itself causes churn.

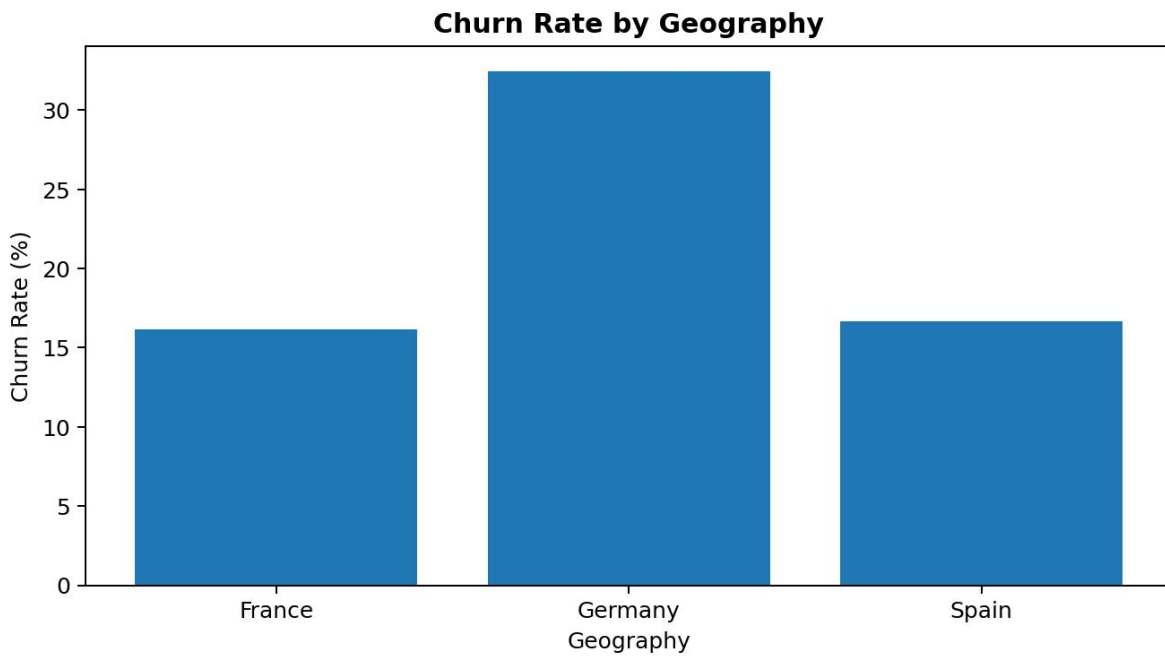


Figure 2. Observed churn rate by geography.

5.3 Age-Based Churn Patterns

Age segmentation reveals substantial variation in observed churn. The 46–60 group has the highest churn rate in this dataset, at 51.12%. The <30 group has the lowest observed churn rate at 7.56%.

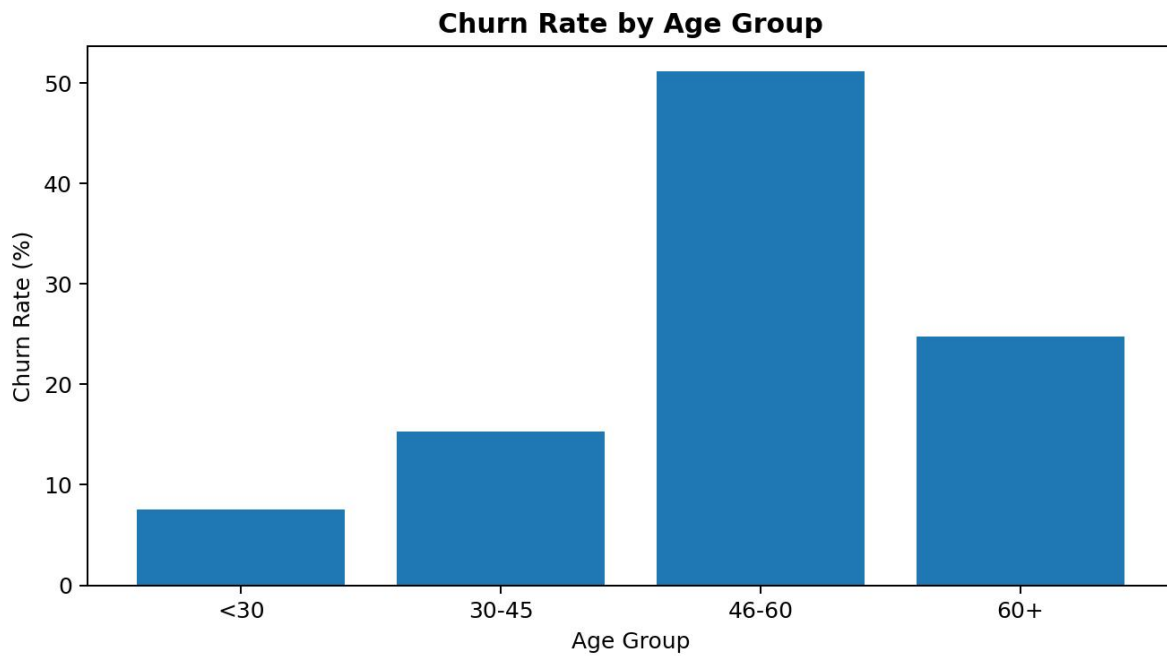


Figure 3. Observed churn rate by age group.

5.4 Tenure-Based Churn Patterns

Tenure differences are smaller than the geographic and age differences. The Long-term group has an observed churn rate of 21.30%, while the Established group has the lowest observed rate at 18.87%. The pattern suggests that relationship duration alone is not sufficient to explain churn.

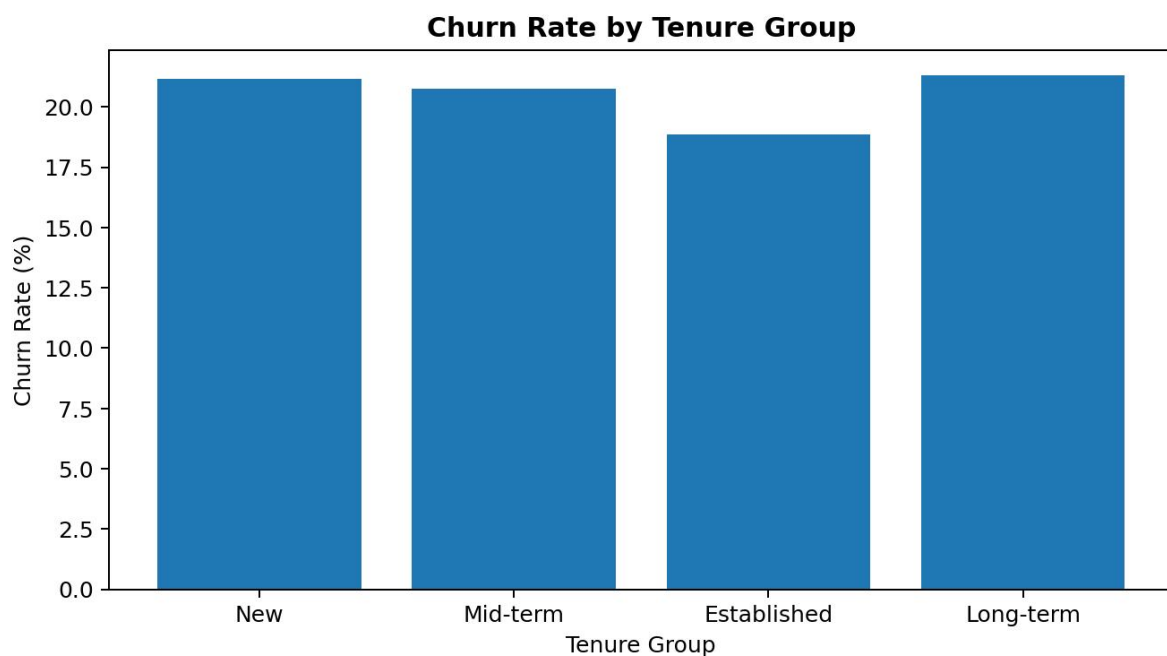


Figure 4. Observed churn rate by tenure group.

5.5 Credit Score and Financial Segments

Churn varies across credit-score bands, although the differences are comparatively moderate. The Poor credit-score band has an observed churn rate of 22.02%. Balance segmentation shows a larger spread: the Low Balance segment has a churn rate of 34.67%, while Zero Balance customers have a lower observed churn rate of 13.82%.

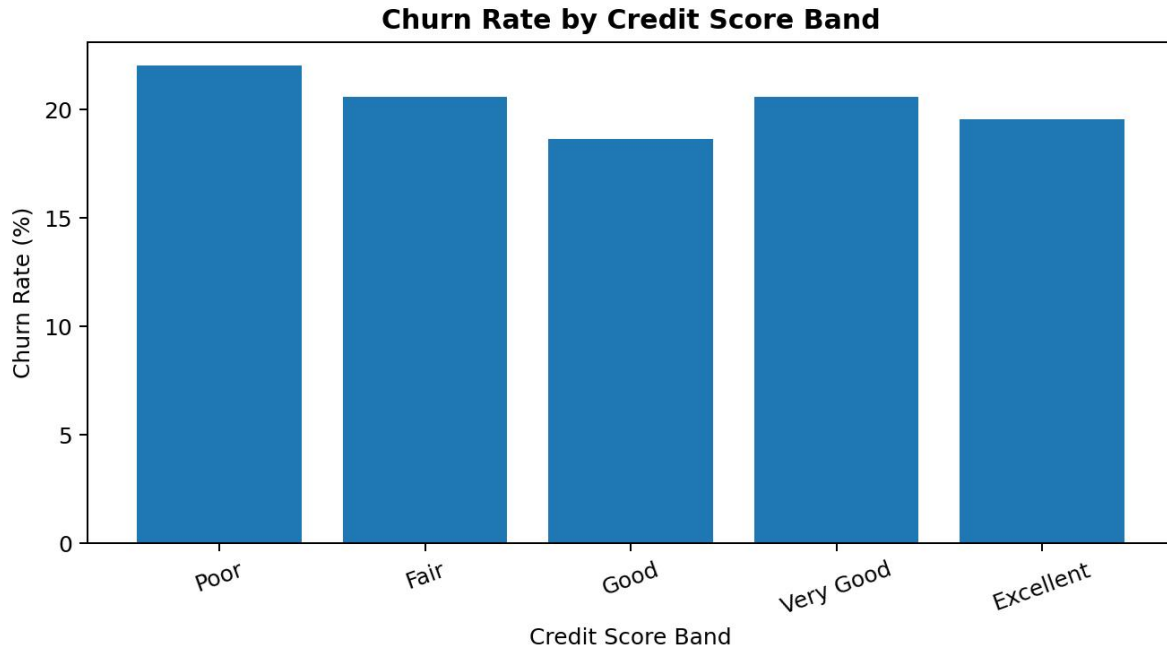


Figure 5. Observed churn rate by credit-score band.

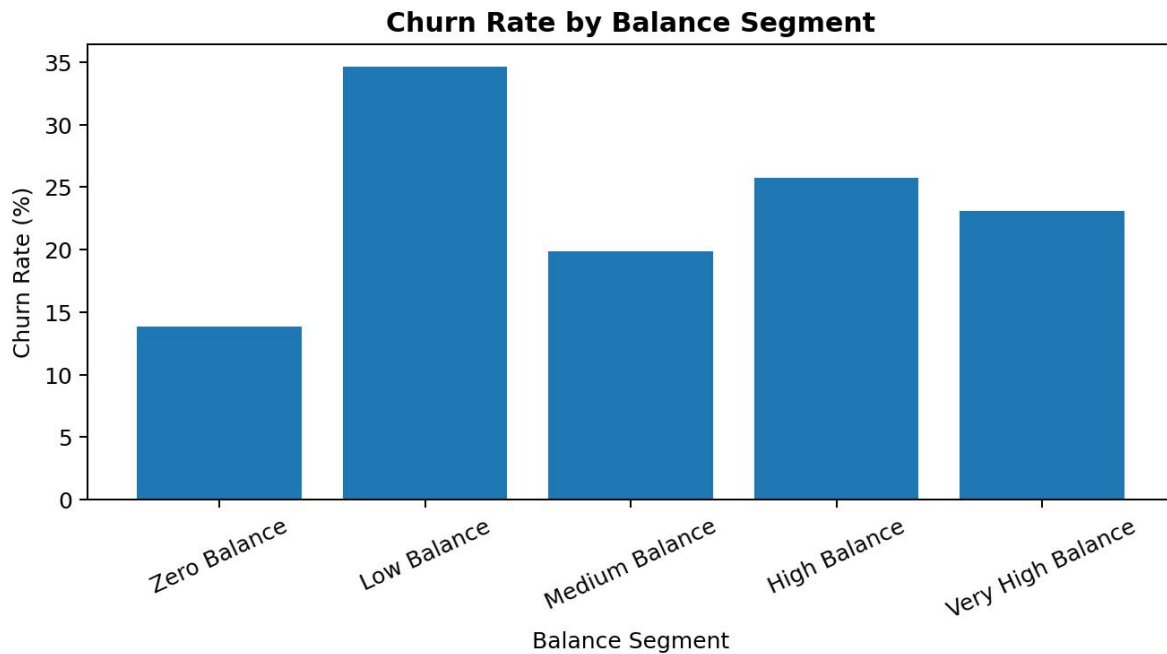


Figure 6. Observed churn rate by balance segment.

5.6 Product Usage

The number of products shows a strong descriptive difference in this dataset. Customers with three products have an observed churn rate of 82.71%, while customers with two products have a much lower rate of 7.58%. The four-product group has a 100% observed churn rate in the supplied data; however, this group is small, so the result should be interpreted cautiously rather than treated as a general banking rule.

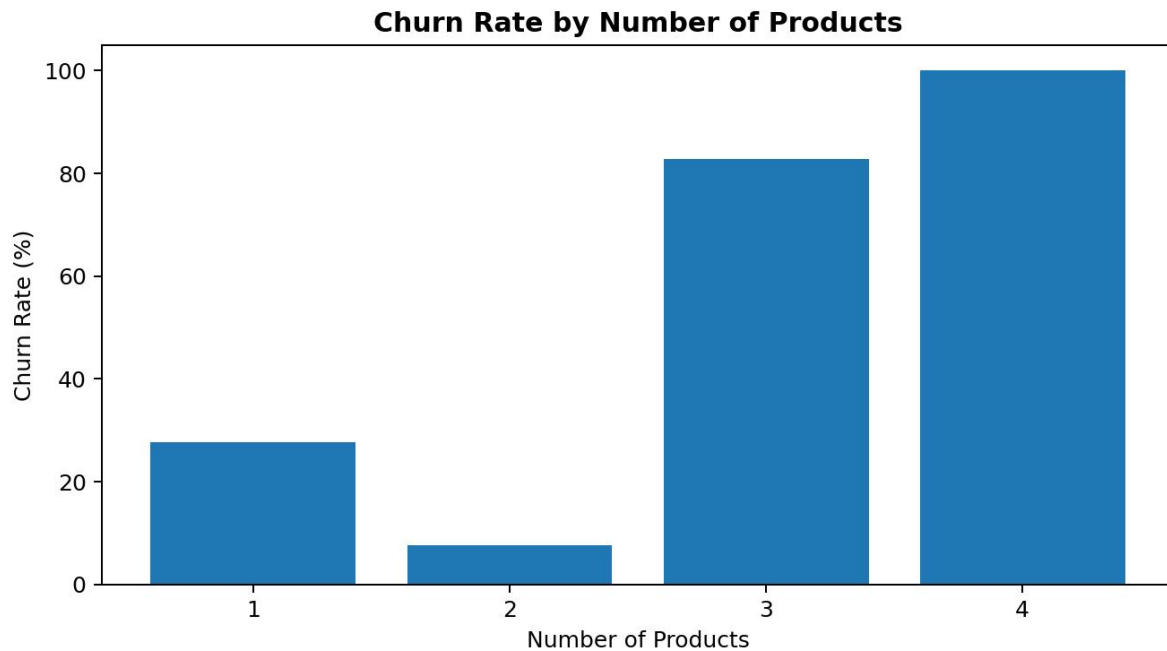


Figure 7. Observed churn rate by number of banking products.

5.7 Customer Engagement

Engagement is one of the clearest descriptive differences in the dataset. Inactive customers show an observed churn rate of 26.85%, compared with 14.27% for active customers. This pattern suggests that customer engagement should be considered when designing retention monitoring.

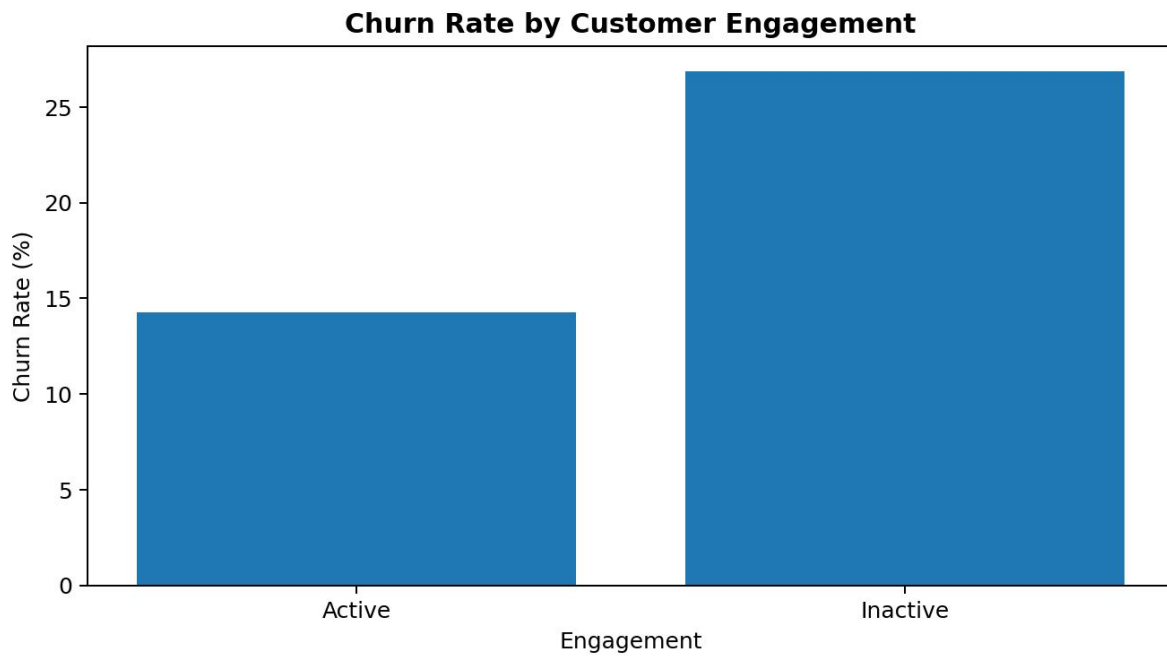


Figure 8. Observed churn rate for active and inactive customers.

5.8 Geography × Age Analysis

The combined geography-by-age heatmap demonstrates how churn can vary across two dimensions simultaneously. This type of drill-down is useful because an overall geographic average can hide differences between age groups.

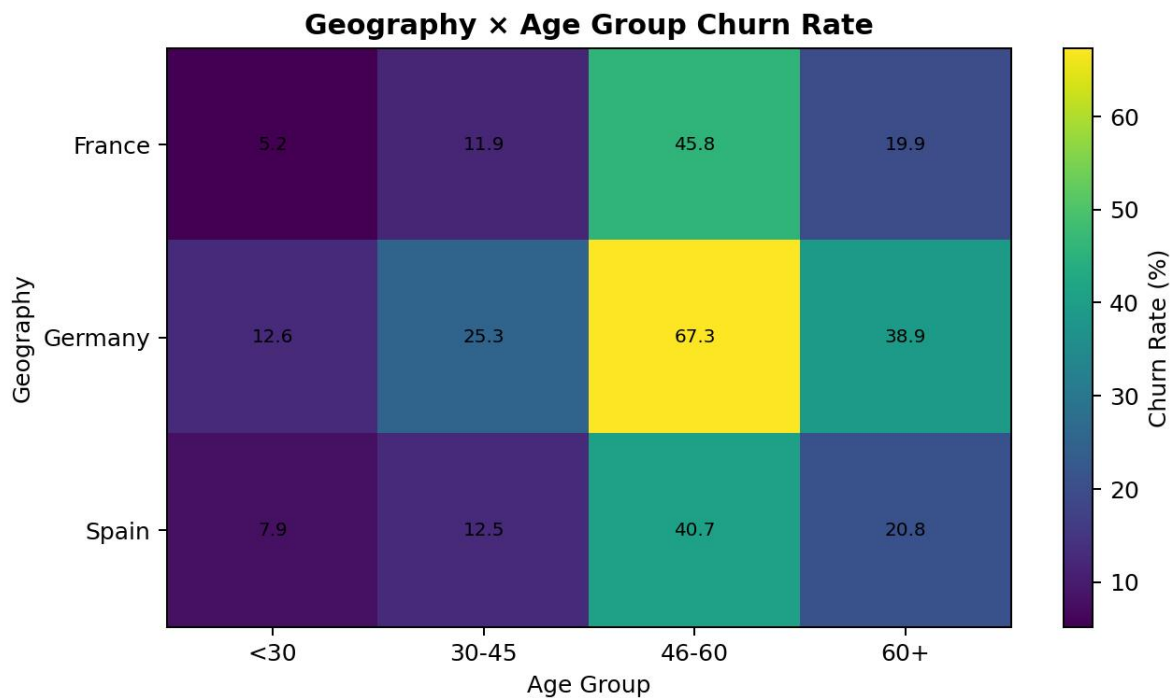


Figure 9. Churn rate (%) across geography and age groups.

5.9 High-Value Customer Analysis

Using the project definition based on median Balance (97,198.54) and median EstimatedSalary (100,193.91), 2,516 customers are classified as High Value. Their observed churn rate is 25.08%, compared with 18.79% among Standard Value customers. Because high-value customers can represent greater financial importance, this segment can be prioritized for deeper retention investigation.

6. Discussion

The findings indicate that churn is heterogeneous across the customer base. Geography, age, engagement and product usage produce particularly visible differences, while tenure and credit-score bands show more moderate variation. The results support the use of segment-level dashboards because aggregate churn rates alone do not reveal which customer groups require attention.

The strong difference between active and inactive customers is especially relevant from a business perspective. Inactivity can be treated as a monitoring signal for retention programs. However, because the dataset is cross-sectional, the observed association should not be interpreted as proof that inactivity causes churn.

The product-count pattern also requires careful interpretation. The very high churn rates for customers with three or four products may reflect a small number of observations or specific customer circumstances. Segment size should therefore be reviewed before operational decisions are made.

7. Business Recommendations

- Develop targeted retention campaigns for customer groups with elevated observed churn rather than applying one strategy to the full portfolio.
- Monitor inactive customers as an early-warning segment and test engagement-focused interventions.
- Prioritize high-value customers for retention review because their churn may have greater potential business impact.
- Investigate the high churn observed in Germany using additional operational, product and service-quality data before assigning a geographic cause.
- Review customers with unusually high product counts and investigate whether product complexity, service issues or customer circumstances contribute to churn.
- Use age and geography together for more precise customer profiling rather than relying on a single demographic dimension.
- Track segment-level churn over time so that retention teams can distinguish persistent patterns from temporary changes.

8. Limitations

- The analysis is observational and does not establish causal relationships.
- The dataset represents a single customer-level snapshot, limiting analysis of churn timing and customer lifecycle dynamics.
- The engineered High Value segment is a portfolio-analysis definition based on dataset medians, not a formal customer lifetime value model.
- Some segments, particularly customers with four products, may contain relatively few observations; extreme rates should therefore be interpreted cautiously.

- The project does not include customer interaction logs, transaction sequences, service complaints, marketing exposure or time-varying behavioral features.
- The Streamlit application is designed for descriptive and diagnostic analytics rather than production churn prediction.

9. Conclusion

This study developed an interpretable customer segmentation and churn analytics framework for 10,000 European banking customer records. The analysis identified an overall churn rate of 20.37% and demonstrated meaningful differences across geographic, demographic, financial, product and engagement segments. Germany and the 46–60 age group showed particularly high observed churn, while inactive customers displayed substantially higher churn than active customers. The analysis also identified a high-value segment that can be monitored separately for retention purposes.

The main practical contribution is an interactive Streamlit dashboard that converts these analytical results into a usable exploration tool. The dashboard combines dynamic KPIs, filters, visualizations, segment analysis and filtered-data export. Future work could extend the project with time-dependent behavioral data, formal clustering, supervised churn prediction, model explainability and controlled retention experiments.

10. References

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Appendix A. Key Dataset Summary

Metric	Value
Customers	10,000
Churned customers	2,037
Retained customers	7,963
Overall churn rate	20.37%
Active customer rate	51.51%
Average balance	76,485.89
High-value customers	2,516

High-value churn rate	25.08%
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Appendix B. Dashboard and Research Workflow

The final implementation follows the workflow: data loading → data validation → cleaning → exploratory analysis → feature engineering → customer segmentation → churn analysis → KPI calculation → interactive visualization → business insights.